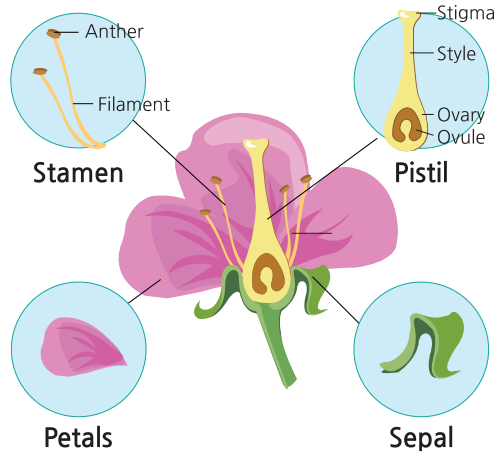
5.2 Sexual reproduction

Seed is produced from a flower by the process of pollination and fertilization. This is known as sexual reproduction. To understand how seeds are formed in a flower, first we need to understand parts of a flower.

ACTIVITY 3

Take a flower. Dissect it longitudinally as shown in the figure and find the parts inside the flower. Can you identify the male reproductive part, androecium (stamen, filament and pollen sac)? Carefully observe the female reproductive part, gynoecium (ovary, style and stigma). If they are not seen clearly, gently pluck off the sepals and petals. Make a drawing of the parts and arrange them in your notebook.



L.S. of a flower showing its parts

5.2.1 Parts of flower

Let us compare few buds and opened flowers of **Hibiscus and Datura.** Observe bud and opened flower of **Hibiscus** and **Datura.** We can tabulate the characteristics of Hibiscus and Datura flowers as below.

Hibiscus flower	
Bud	**Opened flower**
Green colour	Bright colour
Sepals	Petals
Dissected Hibiscus flower	
Bud	**Opened flower**
Curled petals	Expanded petals
Small tube with yellow lobes - Anthers	Expanded tube with yellow lobes -Anthers
	

Datura flower	
Bud	**Opened flower**
Green colour	White colour
Sepals	Petals
Dissected Datura flower	
Bud	**Opened flower**
Curled petals	Expanded petals
Small yellow lobes-Anthers	Expanded yellow lobes-Anthers
	

In a bud, we can see a green colour, leaf like structure which cover the whole bud or flower. Each of these green leaf like structure present as an outermost layer is called as **sepal**. This outer most ring of sepals is known as **calyx.**

Petals are the largest part of flowers. They are often attractive, brightly coloured, sometimes sweet scented and attract the insects. This ring of **petals** together is called **corolla**.